

EARYFOUNDERS LAB

Database Assignment

For this assignment, you will design and build a small MySQL database that reflects a real-world scenario of your choice (for example: a library catalogue, a school club roster, a simple inventory system, a movie collection, etc.). Start by imagining an everyday context where you need to store information in tables, then follow these steps:

Begin by writing a clear “scenario description” in a few sentences. Explain what your database will model and why it’s useful. For instance: “I’m building a bookstore database to track available titles, authors, prices, and stock levels so that the store owner can quickly find which books need restocking.”

Next, translate your scenario into a database schema. In one SQL script file, write the commands to create your database (named whatever you like) and at least one table. Define a primary key column (e.g., id or book_id), give each column an appropriate data type (e.g., VARCHAR, INT, DATE), and include at least five columns that capture different attributes of your chosen entities.

Once your table(s) exist, populate each table with a minimum of ten meaningful rows. Make sure your sample data reflects realistic values and variety: if you’re modeling students, vary the ages or class sections; if you’re modeling products, vary prices and stock counts.

After your data is in place, demonstrate that your database works by writing and running at least three different SELECT queries that answer questions about your data. For example, you might select all items below a certain price, count the number of entries meeting some condition, or sort the results by date or name.

Finally, package your work as follows:

1. A single .sql script file containing everything—from CREATE DATABASE through all CREATE TABLE and INSERT statements—so that anyone can recreate your database from scratch.
2. A short text file (or Word/PDF) with your scenario description and the three SELECT queries you ran along with their output (you can copy-and-paste the query results).

When you submit, I'll look for a clear scenario, a correct and complete SQL script, at least ten rows of well-chosen sample data, and at least three working queries that demonstrate you can extract useful information. This will help you practice designing schemas, writing DDL and DML statements, and querying your own data. Good luck—and have fun modeling something you care about!